Sina Sajadmanesh

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Summary

Senior AI Engineer specializing in multimodal foundation models, with a strong focus on translating research into high-performance, production-ready systems. Proven track record in building scalable model training pipelines and optimizing inference across heterogeneous hardware.

Education

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering — Aug 2023

Sharif University of Technology, MSc in Information Technology Engineering — Sept 2016

University of Isfahan, BSc in Computer Software Engineering — Feb 2014

Experience

Sony AI, Senior AI Engineer – Zurich, Switzerland — Jan 2026 – present

- Optimized and deployed ML models on Qualcomm edge devices using QAIRT / SNPE / QNN, extending official support to modern LLMs and VLMs
- Architected and maintained a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon)
- Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements

Sony AI, AI Engineer – Zurich, Switzerland — Oct 2023 – Dec 2025

- Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS'25 spotlight paper
- Led end-to-end optimization of Sony's Argus VFM inference stack, resulting in up to 61× speedup across 17 vision tasks
- Developed 7 computer vision models, including object detection, depth estimation, super-resolution, and OCR, achieving SOTA or competitive results
- Architected and implemented a foundational MTL framework for Sony's Argus VFM, supporting 17 vision tasks, enabling CVPR'25 publication

Idiap Research Institute, Research Assistant – Martigny, Switzerland — May 2019 – Aug 2023

- Worked on differentially private machine learning with graph neural networks
- 3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations
- 7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter
- 6 open-source projects with 130+ stars on GitHub
- 1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy
- Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe

Brave Software, Research Intern – Remote — Mar 2022 – May 2022

- Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite

Skills and Expertise

Foundation & Generative Models: Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: TensorRT · ONNX · Qualcomm QNN / SNPE / QAIRT · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

Machine Learning Frameworks: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMCV, MMEngine, MMDetection)

Engineering & Tooling: Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

Community and Professional Service

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

Selected Publications

StelLA: Subspace Learning in Low-rank Adaptation using Stiefel Manifold — Dec 2025
Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Lingjuan Lyu
NeurIPS 2025 🌟 Spotlight (top 3%)

Argus: A Compact and Versatile Foundation Model for Vision — June 2025
Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.
CVPR 2025

ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees — Mar 2024
Sina Sajadmanesh, Daniel Gatica-Perez
WSDM 2024

GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation — Aug 2023
Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez
USENIX Security 2023

Locally Private Graph Neural Networks — Nov 2021
Sina Sajadmanesh, Daniel Gatica-Perez
CCS 2021

A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics — May 2020
Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al.
IEEE Internet of Things Journal

Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks — Aug 2019
Sina Sajadmanesh, Sogol Bazargani, Jiawei Zhang, Hamid R. Rabiee
TKDD

Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web — Apr 2017
Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.
WWW 2017